Joe Kanjirathingal

Data Science and Machine Learning
Thrissur, Kerala, India | +91 8075915694 | joepikka5@gmail.com
github.com/Joepikka | LinkedIn | Portfolio

SUMMARY

B.Tech Artificial Intelligence student graduating in March 2027, seeking data science, data analyst and machine learning internships. Fine-tuned and deployed a transformer model for product matching; project experience in Python, SQL, NLP classification, predictive modeling and dashboards, with reproducible evaluation.

EDUCATION

B.Tech in Artificial Intelligence | Sep 2023 - Mar 2027 (expected)

Vidya Academy of Science and Technology, Thrissur, Kerala

TECHNICAL SKILLS

Programming and Data: Python, SQL, pandas, NumPy, data validation
Machine Learning: scikit-learn, PyTorch, Hugging Face Transformers, ONNX, NLP, classification, calibration, macro-F1
Visualization and Applications: GeoPandas, Leaflet, Chart.js, FastAPI, Django, React, TypeScript
Databases and Tools: PostgreSQL, SQLite, Supabase, Git, GitHub, Vercel

SELECTED PROJECTS

QuoteCheck - Supplier Quotation Review and Product Matching | Project

Oct 2026 - Present | Python, PyTorch, FastAPI, React, PostgreSQL, ONNX, Vercel

- Built a quotation-review app with PDF/XLSX extraction, source-linked corrections, exact pricing and exports.
- Fine-tuned MiniLM on 19,835 product pairs; achieved 74.3% validation average precision and 64.7% F1 on a 4,500-pair WDC Products benchmark, versus 43.4% F1 for the logistic baseline.
- Deployed ONNX CPU inference on Vercel with PostgreSQL; verified by 247 automated tests and 3 browser workflows.

IntentDesk - NLP Classification and Evaluation | GitHub

Oct 2026 | Python, scikit-learn, PyTorch, FastAPI, SQLite

- Built an NLP workflow for 150 intent categories using frozen pretrained MPNet embeddings, a downstream classifier and probability calibration.
- Evaluated 4,500 CLINC150 known-intent test requests: 96.24% accuracy and 0.9624 macro-F1, versus 91.96% accuracy for the lexical baseline.
- Created a six-category card-support demo with FastAPI and SQLite review history; retained human review after unsupported-request evaluation failed.

Lilly's POS - Cafe Operations Prototype | Live demo

Sep 2026 - Present | React, TypeScript, PostgreSQL, Supabase

- Built a database-backed cafe prototype with customer, owner and kitchen roles, orders, receipts and sales reports.
- Added diet-filtered guest ordering, pickup/delivery, Google sign-in roles and receipt corrections with reasons.

Kerala Crime Intelligence Map - Data Dashboard | Live demo

Sep 2026 - Present | Python, JavaScript, Leaflet, Chart.js

- Built an interactive dashboard of published Kerala Police statistics with district comparisons, year filters, trend charts, population-adjusted rates and CSV exports.
- Reconciled source totals and distinguished 14-district POCSO coverage from statewide-only categories; documented partial-year 2026 data and population-base limitations.

La Liga - Predictive Modeling and Evaluation | Project

Oct 2026 | Python, PyTorch, NumPy, scikit-learn, SQLite

- Prepared historical football data with source hashes, reconciliation checks and chronological train/test splits.
- Built a leak-free chronological evaluation against a frozen baseline on 86 held-out fixtures; reported that the richer neural model did not improve on it (47.67% vs 50.00% accuracy).

TRAINING

Python Full-Stack Development | SMEClabs, Kochi

Jun 2026 - Present | Expected completion: Dec 2026

Six-month programme covering Python, SQL, Django, HTML/CSS and JavaScript.